Replacement and Reputation

Navin Kartik Elliot Lipnowksi Harry Pei

Motivation

Primary instrument to control elected officials: replacement

Replacement serves two roles:

1. Provide incentives
2. Selection

Other settings: organizational leaders, bureaucrats, etc.

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Our question: Does replacement lead to good outcomes in the long run?

This paper

A stylized model of accountability with moral hazard and adverse selection

- Reputational model: long-lived politicians; short-lived voters
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Takeaways

- Precisely when moral hazard can be well controlled, norms (eqm selection) are critical
  $\rightarrow$ bad norms can lead to bad long-run outcomes
- Only selection assures good long-run outcomes
- Novel source of long-run inefficiencies

Related Literature

Political accountability with selection

- Banks & Sundaram (1993); Fearon (1999); Myerson (2006); Anesi & Buisseret (2022)

Reputation with imperfect monitoring

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Reputation with exogenous replacement

- Mailath & Samuelson (2001); Tadelis (2002); Ekmekci, Gossner, Wilson (2012)

Reputation with competition between long-lived players

- Hörner (2002); Atakan & Ekmekci (2015); Deb & Fanning (2024)

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Time: $t = 0, 1, \ldots$

Pool of infinitely many identical long-lived politicians

Sequence of short-lived voters, one in each period

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In period 0, one politician is exogenously the incumbent

Incumbent chooses effort $a_0 \in \{0, 1\}$ and generates public signal $s_0 \sim f(\cdot|a_0) \in \Delta S$,

- S is finite, $f(\cdot|0) \neq f(\cdot|1)$, and both pmfs have full support

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Period $t \geq 1$:

- Current voter decides whether to replace the incumbent with random new draw from pool
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Replacements are public; once a politician is replaced, he never returns

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- Opportunistic type's stage payoff is

$$u_s = \begin{cases} 0 & \text{if not in office in period } s \\ u(a_s) & \text{if in office in period } s, \end{cases}$$

where $u(0) \equiv 1 > u(1) \equiv 1 - \kappa > 0$. ($\kappa$ is effort cost)

Discount factor δ

Solution Concept

Equilibrium: symmetric weak PBE, i.e., weak PBE in which

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An equilibrium exists

why weak PBE

Main Result

(Eventual) First Best

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Note: negative result also under weaker notions of eventual first best

Equivalence Theorem

If an eqm attains FB, it has no learning; so FB must also be attainable absent good type

Intuitively, this requires conjunction of

low-enough effort cost; sufficiently-informative monitoring; enough patience

Formally on next slide $\rightarrow$ Condition FB-I

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So, always some eqm that attains Eventual FB

But there is a tension between

- Eventual FB in *all* eqa
- FB (or even FB in first period) in *some* eqm

Condition FB-I

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There is a vector $(v(s))_{s \in S} \in [0,1]^S$ such that

$$(1-\delta)(1-\kappa) + \delta \sum_{s \in S} f(s|1) v(s) \geq (1-\delta)1 + \delta \sum_{s \in S} f(s|0) v(s) \qquad (\mathrm{IC_{FB}})$$

and

$$(1-\delta)(1-\kappa) + \delta \sum_{s \in S} f(s|1) v(s) \geq \max_{s \in S} v(s). \qquad (\mathrm{PK_{FB}})$$

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Holds if and only if

- κ is small enough; and
- f is sufficiently informative; and
- δ is large enough

What Prevents Good Long-Run Outcomes?

Theorem says some eqa do not attain Eventual FB when FB-I holds

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Let $\pi(h^t)$ denote the incumbent's reputation at history h^t. (Probability of good type.)

Proposition

Consider any equilibrium that does not attain Eventual FB.

There is a positive-prob history h^t at which
$\pi(h^t) > \pi_0$ and yet the incumbent is replaced with positive prob.

$\Longrightarrow$ replacement despite positive track record
($\because$ correctly expect lower effort than from new incumbent)

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Contrast with the reason for failure of Eventual FB in Myerson (2006)

- There, replacement cost is too high and there is never replacement

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In every eqm:

1. Every opportunistic type is eventually replaced with probability 1.
2. If FB-I fails, every good type is retained forever with positive probability.

Hence, if FB-I fails, then with probability 1, some good type is never replaced.

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Logic for the first part is related to CMS (2004), "impermanent reputations"

Proofs Snippets

Plan

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The following are equivalent:

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I will comment on:

- not (1) $\implies$ not (3)
- (1) $\implies$ (3)

Omit:

- (1) $\implies$ (2) is clear
- (2) [hence (4)] $\implies$ (1)
- (1) $\implies$ (4)

Equilibria that Fail Eventual FB

Why is it not easy to construct eqa that fail Eventual FB?

Naive attempt:

- opportunistic incumbent never works
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Solution when Condition FB-I holds: An eqm in which

- there isn't too much learning
- partial shirking in the first period (and no shirking thereafter, on path)
- in every period the incumbent may be replaced (probabilistic, only after "bad" signals)

Hence Eventual FB violated

▸ Construction

FB-I Fails $\implies$ Eventual FB in All Equilibria

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  - Loose intuition: If $\sup$[on-path beliefs] < 1, then incumbent would have to be working when near that $\sup$, contrary to not FB-I
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- By Doob's upcrossing inequality, every good type has a positive probability of getting stuck at a high-enough belief s.t. never replaced
- So eventually some good type is never replaced

Conclusion

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A model of accountability with moral hazard and adverse selection

- Each politician is either good or opportunistic
- Replacement is instrument to provide incentives and to select good politicians

Takeaways

- Replacement makes long-run good outcomes possible (in some eqm) and, sometimes guarantees it (in all eqa)
- Tension between that guarantee and the possibility of good outcomes in all periods
- Source of long-run inefficiencies is replacement despite positive track record

Appendix

Condition FB-I Fails $\implies \mathbb{P}(a_0 = 1) < 1$ in every eqm

If $\mathbb{P}(a_0 = 1) = 1$, then the opportunistic type must be incentivized to play $a_0 = 1$

(Back)

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If $\mathbb{P}(a_0 = 1) = 1$, then the opportunistic type must be incentivized to play $a_0 = 1$ (Back)

Consider any $\varepsilon > 0$ small enough.

Because FB-I fails, there exists $s_0 \in S$ such that $V(s_0) \geq V(\emptyset) + \varepsilon$.

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- Voter optimality $\implies$ the opportunistic type must again be incentivized to play $a_1 = 1$
- There exists $s_1 \in S$ such that $V(s_0, s_1) \geq V(s_0) + \varepsilon$.

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Iterating, for any $t \geq 0$, there is a pos-prob history $(s_0, \ldots, s_t)$ s.t. $V(s_0, \ldots, s_t) \geq V(\emptyset) + t\varepsilon$

Contradiction, as $V(\cdot) \leq 1$

FB-I $\implies$ Equilibria that Attain FB

Consider the following strategy profile

- Incumbent always exerts effort on the eqm path.
- Voter t replaces the incumbent if and only signal s_{t-1} has a LR below some threshold.
- Off path, voter never retains incumbent and incumbent always shirks.

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- Off path, voter never retains incumbent and incumbent always shirks.

We show that FB-I $\implies$ there is a threshold s.t. this profile is an eqm.

- Only thing to check is incumbent's incentive

Lemma

Assume Condition FB-I. There exist $\overline{v} > 0$ and $x \in (0, 1]$ such that (IC_{FB}) holds and

$$(1-\delta)(1-\kappa) + \delta \sum_{s \in S} f(s|1) v(s) = \overline{v},$$

with $v(s) = \overline{v} \cdot \mathbf{1}\left\{\frac{f(s|1)}{f(s|0)} \geq x\right\}$

Incentive Lemma when FB-I Fails

Lemma

If Condition FB-I fails, then $\exists \varepsilon > 0$ s.t. $\forall \left(v(s)\right)_{s \in S} \in [0,1]^S$ that satisfies ($\mathsf{IC}_{\mathsf{FB}}$),

$$\max_{s \in S} v(s) \geq (1-\delta)(1-\kappa) + \delta \sum_{s \in S} f(s|1) v(s) + \varepsilon.$$

Strategies & Solution Concept

Equilibrium: symmetric weak PBE, i.e., weak PBE in which (Back)

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- Strengthens one direction of main result
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Proposition

A symmetric PBE exists.

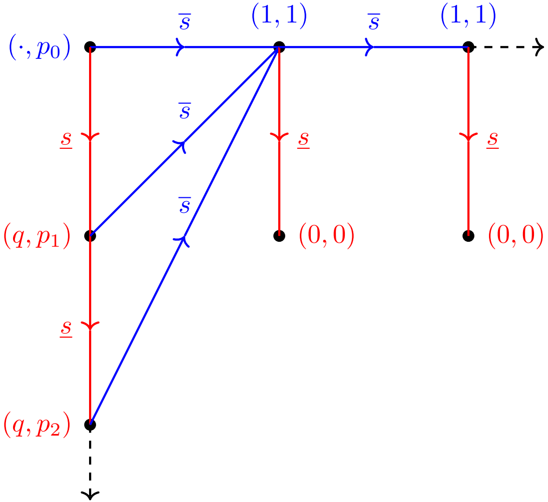
Idea: auxiliary game with just one politician; a voter's payoff on replacement is $\pi_0 + (1 - \pi_0)\sigma_P(\emptyset)$

FB-I $\Longrightarrow$ Equilibria that Fail Eventual FB

Construction with two signals, $S = \{\underline{s}, \overline{s}\}$.

FB-I $\Longrightarrow$ Equilibria that Fail Eventual FB

Construction with two signals, $S = \{\underline{s}, \overline{s}\}$. The first number in the vector is the prob of retention, and the second number is the prob of choosing $a = 1$. It holds that $0 < p_0 < p_1 < \cdots < 1$ and $q \in [0, 1)$.



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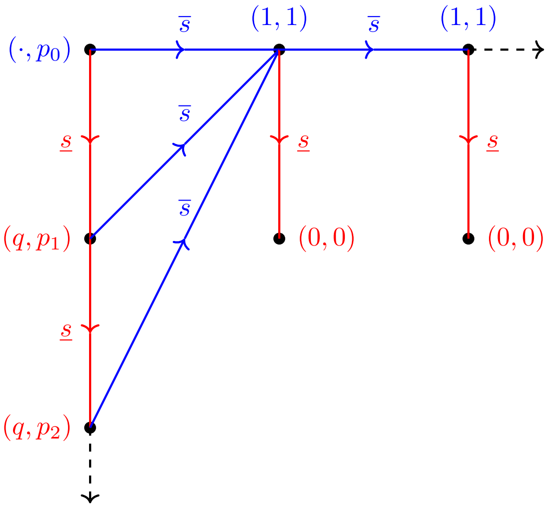

(Back)

Caveat:
use weak PBE to sustain $(0,0)$ profiles